

Question number	Answer	Additional guidance	Mark
5(a)(i)	An explanation that includes three of the following points: - because ATP is the source of energy for {plants / all living organisms} (1) - because light energy cannot be used (directly) (1) - ATP is needed in the {light-independent reactions/ Calvin cycle} to convert GP into GALP (1)	ACCEPT ATP is the {usable source of energy / energy currency}	(2)

Question number	Answer	Additional guidance	Mark
5(a)(ii)	An answer that includes the following points: (ADP / adenosine diphosphate) and (inorganic) {Pi / phosphate / PO_4^{3-} (ion)} (1) - hydrolysis (1)	ACCEPT adp DO NOT ACCEPT P / phosphorus ACCEPT dephosphorylation	(2)

Question number	Answer	Additional guidance	Mark
5(a)(iii)	An explanation that includes three of the following points: • to {release / excite} electrons (from chlorophyll) (1) • so that <u>electrons</u> can be used in {chemiosmosis / (photo)phosphorylation} (1) • photolysis to {replace electrons lost by chlorophyll / provide protons for formation of reduced NADP} (1) • to produce ATP and reduced NADP for the {light-independent reactions / Calvin Cycle} (1)		(3)

Question number	Answer	Mark
5(b)(i)	The only correct answer is D. <i>A is incorrect because ATP is not available to light-dependent reactions in cyclic photophosphorylation</i> <i>B is incorrect because ATP is not available to light-dependent reactions in cyclic photophosphorylation</i> <i>C is incorrect because oxidised NADP is not produced</i>	(1)

Question number	Answer	Additional guidance	Mark
5(b)(ii)	An answer that includes the following points: • C and O from carbon dioxide (1) • H from water (1)	ACCEPT CO₂ ACCEPT H₂O DO NOT ACCEPT O from water	(2)

Question number	Answer					Mark
5(b)(iii)						(3)
	New biological molecule	Nitrates	Phosphates	Both nitrates and phosphates	Neither phosphates nor nitrates	
	protein	X				
	RNA			X		
	triglyceride				X	